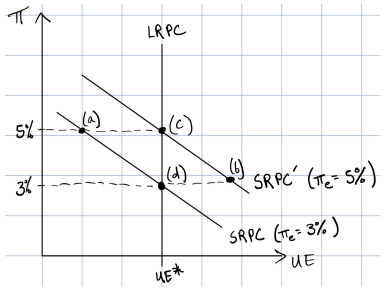
Ch. 36 Solutions

Principles of Macroeconomics

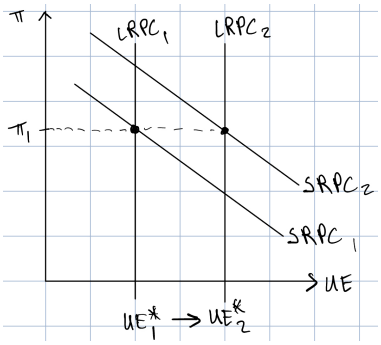
May 5, 2026

Problems and Applications

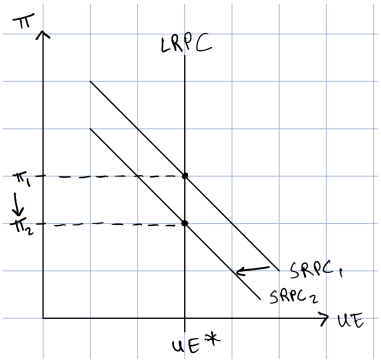
1.



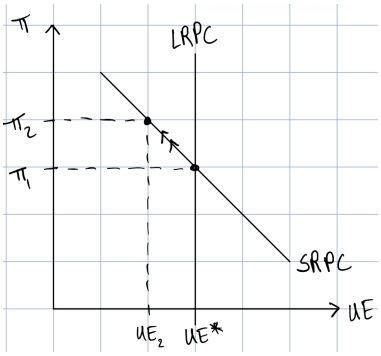
2. (a) If the natural rate of unemployment increases, the long-run Phillips curve shifts right because the long run Phillips curve is vertical at the natural rate of unemployment. Typically, a development that increases the natural rate of unemployment will shift LRAS and SRAS to the left. That's because a higher unemployment rate means fewer people are working, and when fewer people are working, the economy produces less. When SRAS shifts left, it causes the short-run Phillips curve to shift right.



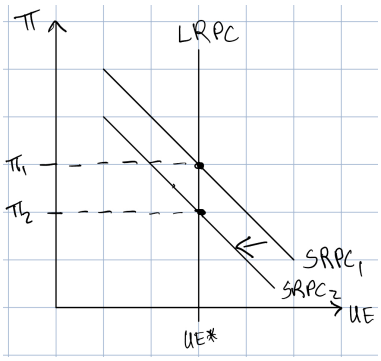
(b) A decline in the price of oil is a positive supply shock, so the SRAS curve shifts to the right. In response to a positive supply shock, the short-run Phillips curve shifts left.



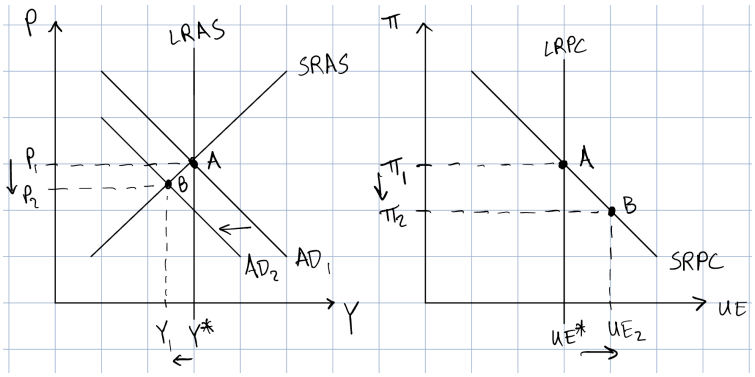
(c) A rise in government spending shifts the AD curve to the right. That decreases unemployment and increases inflation, so there's a movement up and to the left along the short-run Phillips curve.



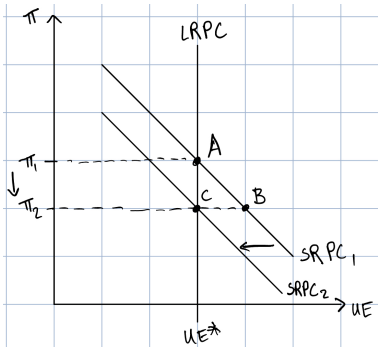
(d) When expected inflation declines, the SRAS curve shifts to the right. That causes the short-run Phillips curve to shift to the left.



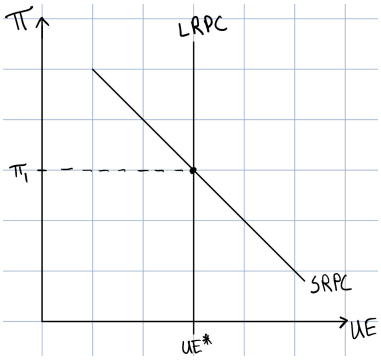
3. (a) In the short run, inflation decreases and unemployment increases.



(b) If expected inflation decreases, then the short-run Phillips curve will shift to the left. After the recession, the economy faces a better set of inflation-unemployment combinations because for any given level of unemployment, the corresponding inflation rate is now lower.

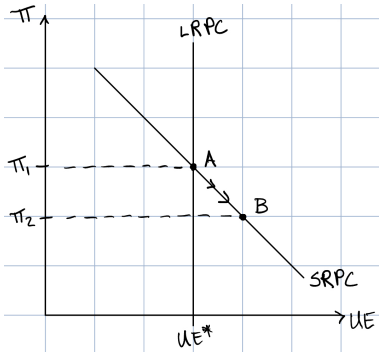


4. (a)

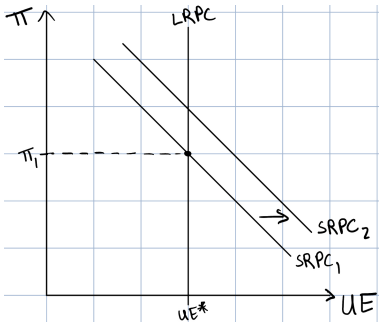


(b) If the AD curve shifts left, the price level and output fall. As a result, inflation decreases, and the unemployment rate increases. That causes a movement down and to the left along the short-run Phillips curve.

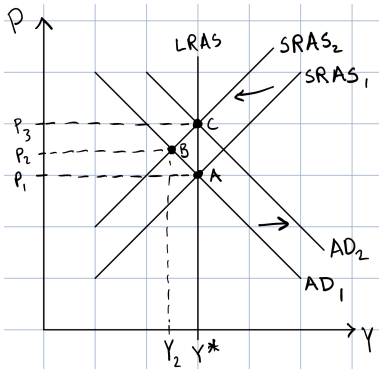
If the Fed undertook expansionary monetary policy, it could shift the AD curve back to its original position. Doing so would return the economy to its original inflation rate and unemployment rate.



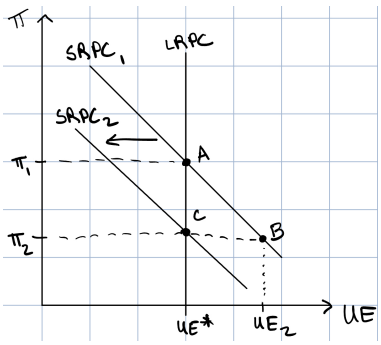
(c) An increase in oil prices is a negative supply shock. If the SRAS curve shifts left, the Phillips curve shifts right.



If the Fed undertakes expansionary monetary policy, it can return the economy to its original unemployment rate but it can't return it to its original inflation rate. Expansionary monetary policy will shift AD to the right which will restore full-employment output but increase inflation.

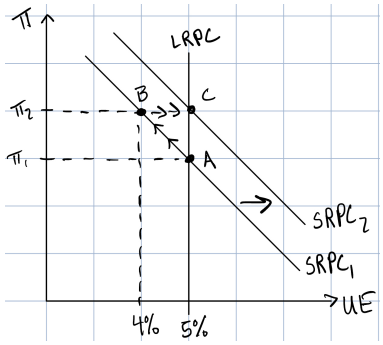


5. Milton is more likely to favor the proposed change. When the Fed slows the rate of money growth, the economy will experience a period of low output and high unemployment while disinflating.

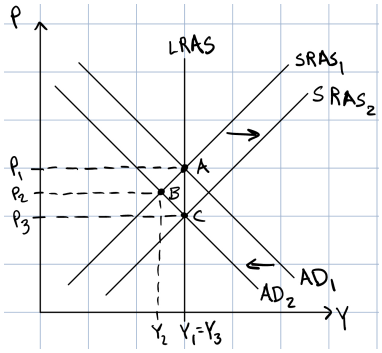


If people change their expectations quickly, that period will be briefer because the Phillips curve will shift left more quickly.

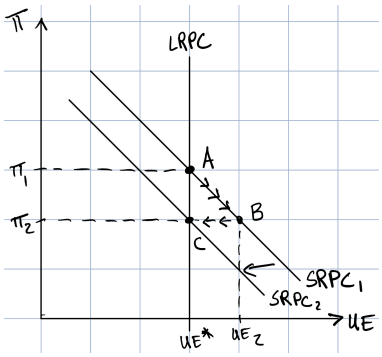
6. The Fed would implement expansionary monetary policy to increase output until unemployment was 4%. Doing so would shift the economy up and to the left on the SRPC which would increase inflation. In the long run, people's inflation expectations would adjust to match the new inflation rate. At that point, the SRPC would shift right until unemployment returned to its natural rate. If the Fed didn't realize its mistake, that process would repeat itself and inflation would increase again. Eventually, the Fed would have to realize that each time it stimulated aggregate demand, the decrease in unemployment was only temporary.



7. (a) The ensuing recession would be less severe. When the Fed pursues contractionary monetary policy, the AD curve shifts left, and the price level falls. If wage contracts have short durations, then firms and works will quickly adjust their expectations to the new price level, and SRAS will shift right more quickly. That restores the economy to its natural level of output and unemployment sooner.

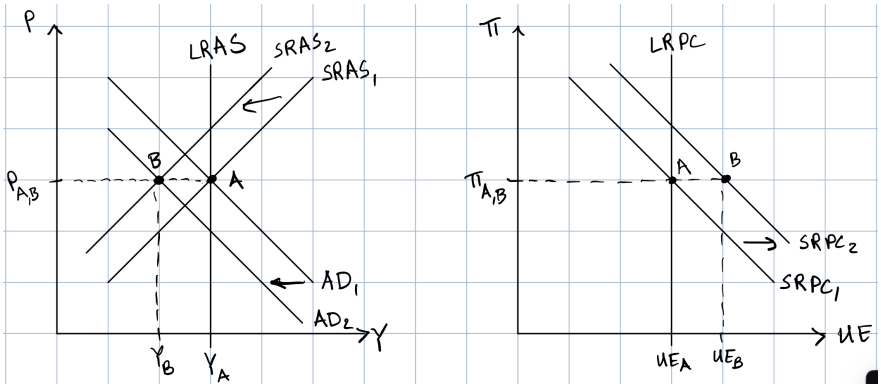


(b) The ensuing recession would be more severe. If people had little confidence in the Fed's determination, they would adjust their inflation expectations more slowly. As a result, it would take the SRPC longer to shift to the left, and the economy would spend longer in a state of lower output and higher unemployment.

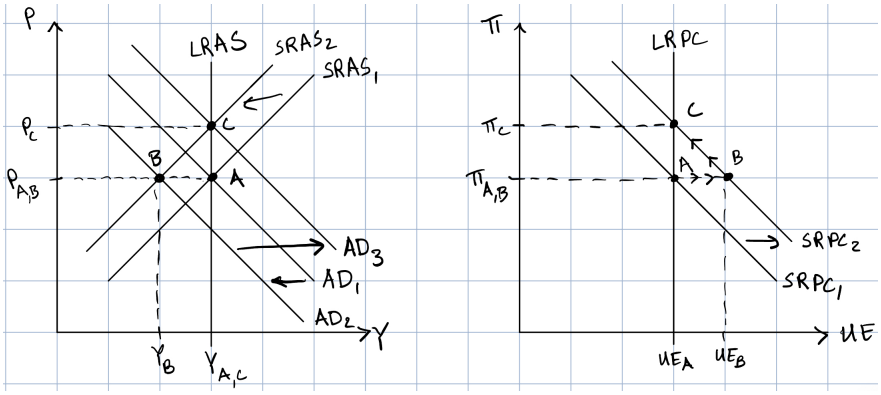


(c) The ensuing recession would be less severe. If people adjust their expectations quickly, the SRPC will shift left more quickly, and the economy will return to its natural level of unemployment and output sooner.

8. (a) Output falls, and unemployment rises. The price level and the inflation rate change ambiguously.



(b) The Fed would implement expansionary monetary policy to shift the AD curve to the right until the economy is back at its natural level of output.



(c) The Fed might not choose to pursue that course of action because it leaves the economy at a permanently higher level of inflation.